SE 4347.004: Database Systems

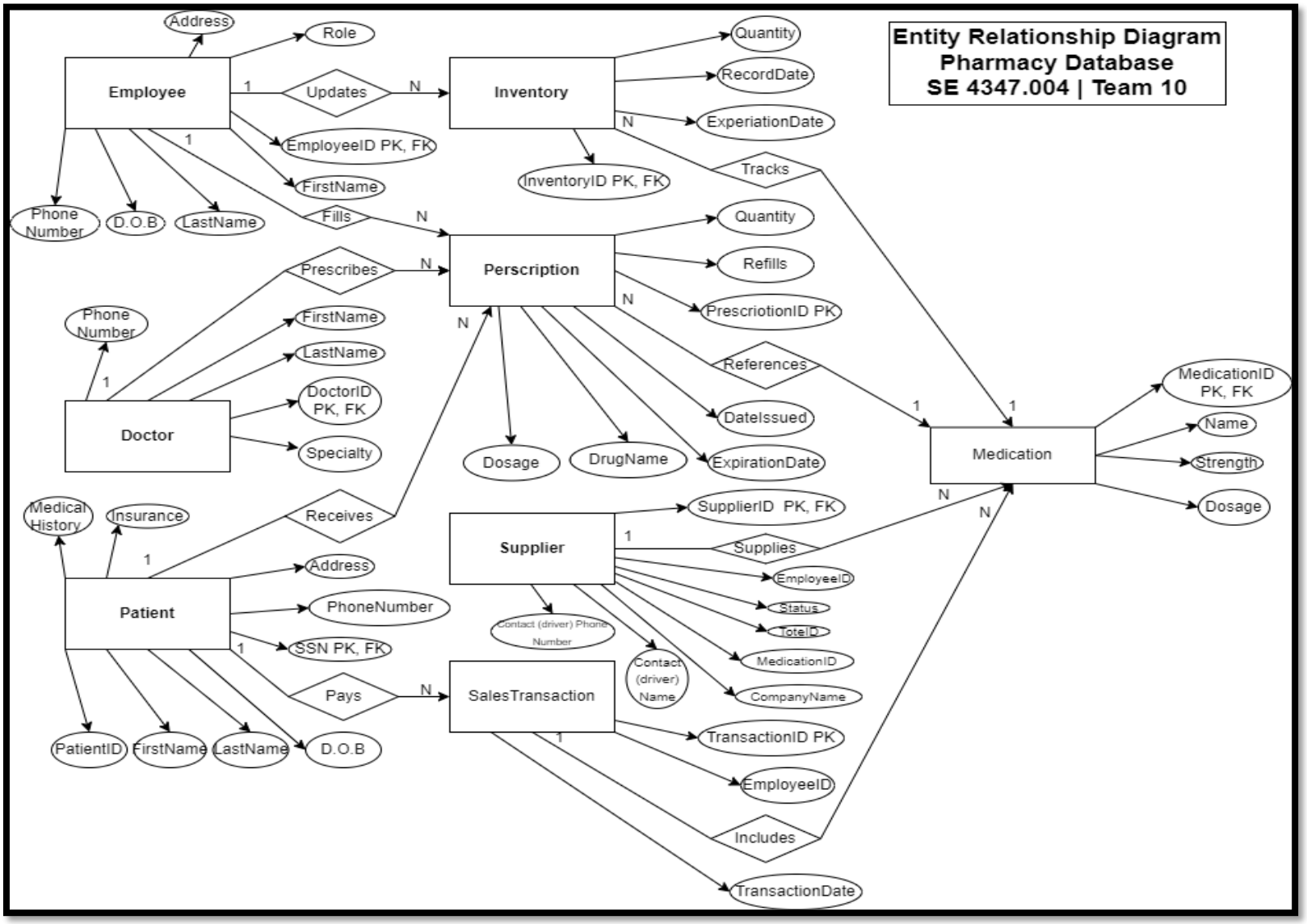
Project Part 1

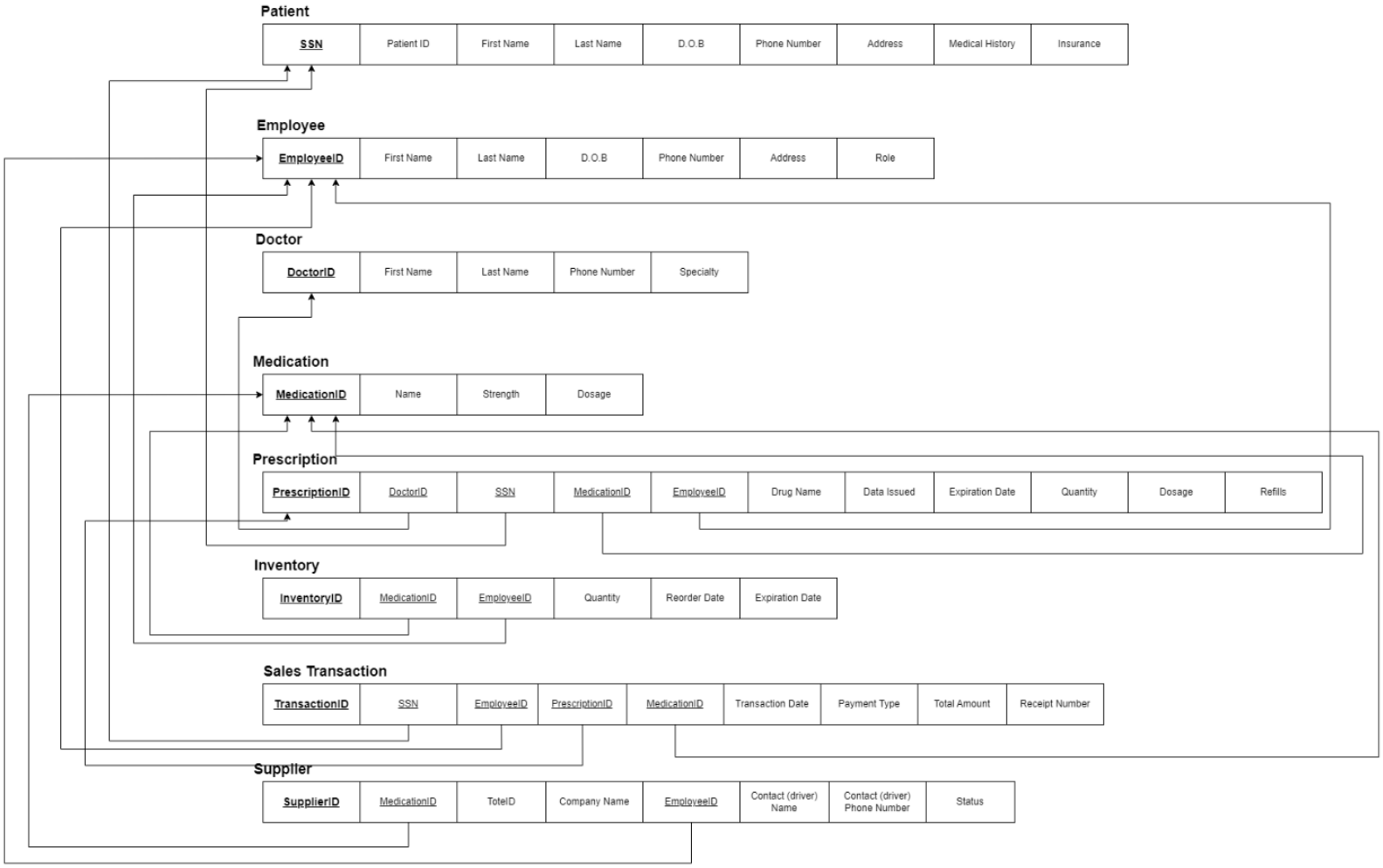
Spring 2026

Team 10

Team Members	Contributions to Document
Eric Carrera	Data Dictionary
Jack Thomas Hedgcoxe	Data Dictionary
Jaimin Kim	Data Dictionary
Mohamad Kiwan	Relational Database Schema
Viet-Long Hoang Nguyen	Entity Relationship Diagram

Entity Relationship Diagram (ERD) & Relational Database Schema (Tool used: Draw.io)





Bold Underlined = Primary Key (PK) | Underlined in FK position = Foreign Key referencing another table | SE 4347.004 · Team 10

Foreign Key Arrows

Each arrow points **FROM** the FK column **TO** the PK it references.

From (FK)	→	To (PK)
PRESCRIPTION.DoctorID	→	DOCTOR.DoctorID
PRESCRIPTION.SSN	→	PATIENT.SSN
PRESCRIPTION.MedicationID	→	MEDICATION.MedicationID
PRESCRIPTION.EmployeeID	→	EMPLOYEE.EmployeeID
INVENTORY.MedicationID	→	MEDICATION.MedicationID
INVENTORY.EmployeeID	→	EMPLOYEE.EmployeeID
SALESTRANSACTION.SSN	→	PATIENT.SSN
SALESTRANSACTION.MedicationID	→	MEDICATION.MedicationID
SALESTRANSACTION.PrescriptionID	→	PRESCRIPTION.PrescriptionID
SALESTRANSACTION.EmployeeID	→	EMPLOYEE.EmployeeID
SUPPLIER.MedicationID	→	MEDICATION.MedicationID
SUPPLIER.EmployeeID	→	EMPLOYEE.EmployeeID

Data Dictionary

Table	Table Description	Primary Key	Super Key	Attributes			Foreign Key
Patient	Pharmacy customer who receives	SSN	(SSN, FullName, D.O.B, Phone)	Attribute	Data Type	Domain	SSN

<table>
<tr><td></td><td>prescriptions and buys medication.</td><td></td><td></td><td>
<table>
<tr><td>SSN</td><td>VARCHAR</td><td>Valid SSN format</td></tr>
<tr><td>PatientID</td><td>INT</td><td>Positive integer</td></tr>
<tr><td>First Name</td><td>VARCHAR</td><td>Alphabetic characters</td></tr>
<tr><td>Last Name</td><td>VARCHAR</td><td>Alphabetic characters</td></tr>
<tr><td>Date of Birth</td><td>DATE</td><td>Calendar date</td></tr>
<tr><td>Phone Number</td><td>VARCHAR</td><td>Valid phone number</td></tr>
<tr><td>Address</td><td>VARCHAR</td><td>Valid address</td></tr>
<tr><td>Medical History</td><td>VARCHAR</td><td>Patient medical history</td></tr>
<tr><td>Insurance</td><td>VARCHAR</td><td>Valid insurance name</td></tr>
</table>
</td><td></td></tr>
<tr><td>Employee</td><td>Staff member at the pharmacy</td><td>Employee ID</td><td>(EmployeeID, FullName, Role)</td><td>
<table>
<tr><td>Attribute</td><td>Data Type</td><td>Domain</td></tr>
<tr><td>EmployeeID</td><td>INT</td><td>Positive integer</td></tr>
<tr><td>First Name</td><td>VARCHAR</td><td>Alphabetic characters</td></tr>
<tr><td>Last Name</td><td>VARCHAR</td><td>Alphabetic characters</td></tr>
</table>
</td><td>Employee ID</td></tr>
</table>

				Date of Birth	DATE	Calendar date	
				Phone Number	VARCHAR	Valid phone number	
				Address	VARCHAR	Valid address	
				Role	VARCHAR	Pharmacist or Technician	
Doctor	Licensed healthcare provider that prescribes medications to patients	DoctorID	(DoctorID, FullName, NPI)	Attribute	Data Type	Domain	DoctorID
				DoctorID	INT	Positive integer	
				First Name	VARCHAR	Alphabetic characters	
				Last Name	VARCHAR	Alphabetic characters	
				Specialty	VARCHAR	Medical specialty	
				Phone Number	VARCHAR	Valid phone number	
				License Number	VARCHAR	Valid medical license	
Medication	Pharma-ceutical drug provided to	MedicationID	(MedicationID, Name, Strength)	Attribute	Data Type	Domain	SupplierID

	patients by pharmacy			Medication ID	INT	Positive integer	
				Name	VARCHAR	Alphabetic characters	
				Strength	VARCHAR	Alphanumeric text	
				Dosage	VARCHAR	Alphanumeric text	
Prescription	Authorization for a patient to receive medication that is issued by a doctor	PrescriptionID	(PrescriptionID, DateIssued, ExpirationDate)	Attribute	Data Type	Domain	DoctorID SSN MedicationID EmployeeID
				Prescription ID	INT	Positive integer	
				DrugName	VARCHAR	Alphabetic characters	
				DateIssued	DATE	Calendar date	
				Expiration Date	DATE	Calendar date	
				Quantity	INT	Positive integer	
				Dosage	VARCHAR	Alphanumeric text	
				Refills	INT	Positive integer	
Inventory	Tracks stock and availability of medication at pharmacy	Inventory ID	(InventoryID, MedicationID)	Attribute	Data Type	Domain	MedicationID EmployeeID
				InventoryID	INT	Positive integer	
				Quantity	INT	Positive integer	
				ReorderDate	DATE	Calendar date	

<table>
<tr><td></td><td></td><td></td><td></td><td><table><tr><td>ExpirationDate</td><td>DATE</td><td>Calendar date</td></tr></table></td><td></td></tr>
<tr><td>Sales Transaction</td><td>Track sales transactions for prescription purchases by patient.</td><td>TransactionID</td><td>(TransactionID, Date)</td><td><table><tr><td>Attribute</td><td>Data Type</td><td>Domain</td></tr><tr><td>TransactionID</td><td>INT</td><td>Positive integer</td></tr><tr><td>EmployeeID</td><td>INT</td><td>Positive Integer</td></tr><tr><td>Transaction Date</td><td>DATE</td><td>Calendar date</td></tr><tr><td>Payment Type</td><td>VARCHAR</td><td>Alphabetic characters</td></tr><tr><td>Total Amount</td><td>VARCHAR</td><td>Alpha-numeric text</td></tr><tr><td>Receipt Number</td><td>INT</td><td>Positive Integer</td></tr></table></td><td>SSN
MedicationID
PrescriptionID
EmployeeID</td></tr>
</table>

<table>
<tr>
<td>Supplier</td>
<td>Warehouse Supplier that delivers medication ordered by the pharmacy.</td>
<td>SupplierID</td>
<td>(SupplierID, Name, Phone)</td>
<td>
<table>
<tr><th>Attribute</th><th>Data Type</th><th>Domain</th></tr>
<tr><td>SupplierID</td><td>INT</td><td>Positive integer</td></tr>
<tr><td>Tote ID</td><td>INT</td><td>Positive integer</td></tr>
<tr><td>MedicationID</td><td>INT</td><td>Positive integer</td></tr>
<tr><td>Company Name</td><td>VARCHAR</td><td>Alphabetic characters</td></tr>
<tr><td>Contact (driver) Name</td><td>VARCHAR</td><td>Alphabetic characters</td></tr>
<tr><td>Contact (driver) phone number</td><td>VARCHAR</td><td>Alpha-numeric text</td></tr>
<tr><td>EmployeeID</td><td>INT</td><td>Positive integer</td></tr>
<tr><td>Status</td><td>VARCHAR</td><td>Alphabetic characters</td></tr>
</table>
</td>
<td>MedicationID
EmployeeID</td>
</tr>
</table>